

Executive Business Report

1. Executive Summary

The business challenge this project set out to solve is a familiar one for any growing retail operation: inventory decisions were being made reactively, based on gut feel and last month's numbers, rather than on a forward-looking view of demand. To address this, four years of order-level sales data (nearly 9,800 transactions across three product categories, seventeen sub-categories, and four U.S. regions) were analyzed to understand how revenue has actually behaved over time, where it is concentrated, and what patterns repeat every year. On top of that analysis, a demand intelligence system was built that does three things: it forecasts sales three months ahead, it automatically flags unusual sales weeks that deserve a closer look, and it groups products into demand segments so that inventory strategy can be tailored rather than one-size-fits-all. The business value is straightforward. Instead of finding out about a demand surge or a slowdown after it has already strained the warehouse or tied up working capital in slow-moving stock, the Supply Chain and Finance teams now have an early-warning and planning tool that turns historical transaction data into forward-looking, segment-specific guidance for purchasing, staffing, and cash allocation decisions.

2. Key Findings from Data Exploration

Technology is the single largest revenue-generating category, narrowly ahead of Furniture and Office Supplies, which are themselves close to one another in total contribution. This tells us the business is not overly dependent on one category, but Technology deserves the most attention when it comes to stock availability, since it carries the most revenue risk

if it runs short.

Regionally, the West is the strongest performer, followed by the East, with the Central region trailing and the South generating noticeably less revenue than the other three. This is a meaningful gap rather than a marginal one, and it raises a natural question for leadership about whether the South is underserved, under-marketed, or simply a smaller market opportunity.

Seasonality is one of the clearest patterns in the data. Sales consistently climb toward the final months of each year, with November and December standing out as the strongest period annually. This is consistent with typical holiday and year-end purchasing behavior and should be treated as a predictable, recurring event rather than a surprise each time it happens.

On shipping, order-to-ship duration is fairly consistent across sub-categories and regions, generally landing in a narrow band of a few days. There is no single region or product line that stands out as a major fulfillment bottleneck, which suggests current logistics performance is stable, though it is worth monitoring as order volumes grow.

Looking at the four-year span as a whole, the business dipped slightly in the second year before entering a sustained growth phase in years three and four, with year four posting the strongest revenue of the period. The overall trajectory is upward, and the growth has been accelerating rather than flattening, which is an encouraging signal but also means the planning system needs to keep pace with a business that is scaling.

3. Three-Month Sales Forecast

Three independent forecasting approaches were tested against held-out recent months to see which one most reliably predicted actual results. The best-performing approach pointed to continued growth over the next three months across the business, broadly consistent with the upward trajectory already observed in the historical data. That said, no forecast is a guarantee, and the models' track record on recent held-out months suggests the actual outcome could reasonably land within roughly fifteen

to twenty percent of the projected figures in either direction. This is a normal and expected level of uncertainty for a three-month retail forecast, not a weakness specific to this exercise.

For inventory planning, the practical takeaway is to treat the forecast as a directional guide for building stock ahead of expected demand, with a buffer built in rather than ordering to the exact predicted number. Categories and regions projected to grow the fastest, discussed further below, warrant earlier and larger purchase orders, while flatter segments can be managed closer to current run-rates.

4. Anomaly Detection Findings

Three unusual sales events stand out as the most important for the business to understand.

The single largest anomaly in four years occurred in a late-March week, when weekly sales spiked to roughly three to four times a typical week's volume, well above anything else recorded that year. This is the kind of spike that is consistent with either a large one-off corporate order or a short, aggressive promotional push. Because it was so far outside the normal range, it is the anomaly most likely to have created real strain on order fulfillment and stock availability at the time, and it is the clearest example of why an early-detection system has direct operational value.

The second pattern worth highlighting is a cluster of very strong weeks in November and early December of the most recent year, each individually among the highest-volume weeks on record. Unlike the March spike, this looks like a seasonal surge rather than a one-off event, which is reassuring: it is predictable, recurring, and something the business can plan capacity around in advance rather than reacting to it after the fact.

The third notable event is a sharp spike in mid-September of the first year, at a volume comparable to the November holiday peaks despite falling well outside the usual holiday season. This is a useful reminder that not every major demand event follows the calendar. It could reflect a back-to-school or office-refresh buying cycle, or a promotional

campaign, but the key business implication is that relying only on holiday-season planning would have missed it entirely, reinforcing the value of ongoing, real-time monitoring rather than a purely seasonal playbook.

5. Product Demand Segmentation

The product portfolio splits into four clear demand segments, and each calls for a different inventory posture.

The High Revenue segment, anchored by copier-type products, generates the highest revenue per order by a wide margin and has shown strong growth, but it is also the most volatile segment in the portfolio. The recommended approach is to maintain higher safety stock on these items so that a sudden order does not result in a stockout on the business's most valuable product line.

The Growing Demand segment, which includes core lines such as chairs, phones, storage, tables, binders, bookcases, and accessories, has shown consistent upward momentum. Procurement for this group should be increased gradually and proactively rather than waiting for stock levels to force a reorder, since under-ordering here risks leaving genuine, growing demand unmet.

The Stable Products segment, covering items such as appliances, paper, furnishings, supplies, and several smaller sub-categories, shows steady, low-volatility demand. These are strong candidates for automated replenishment rules, since their predictability means manual oversight adds little value and mostly adds cost.

The Volatile Products segment, represented here by machines, combines high swings in demand with a flat-to-declining growth trend. This combination is the riskiest to over-invest in. The recommendation is to monitor this segment closely on a short cycle and deliberately avoid building up large stock positions, since both the direction and the size of future demand are the least predictable of any segment.

6. Business Recommendations

Recommendation 1: Build Technology and West/East region inventory ahead of the next quarter. Why it matters: Technology is the top revenue category and the West and East regions are the two strongest markets, and the forecast points to continued growth concentrated in these areas. Expected business impact: Reduces the risk of stockouts in the segments that would cost the business the most in lost revenue if demand outpaces supply.

Recommendation 2: Adopt a segment-specific inventory policy instead of a uniform reorder rule. Why it matters: The analysis shows genuinely different behavior across High Revenue, Growing Demand, Stable, and Volatile products, and treating them identically means either over-investing in slow movers or under-protecting fast movers. Expected business impact: Frees up working capital currently tied up in excess stock of stable and volatile items, while directing that capital toward the products where it will protect revenue.

Recommendation 3: Put an ongoing anomaly-monitoring process in place, rather than only reviewing sales after the fact. Why it matters: The March and September spikes identified in this analysis both occurred outside the predictable holiday season and would not have been caught by seasonal planning alone. Expected business impact: Gives Supply Chain an earlier warning window to expedite stock or adjust staffing when an unusual demand event begins, shortening the gap between "something unusual is happening" and "we have responded to it."

7. Risks and Limitations

The most important limitation to flag is that these forecasts and segment classifications are built entirely on four years of internal transaction history. They do not yet account for external factors that can shift demand independently of past patterns, such as broader economic conditions, competitor pricing moves, supply disruptions upstream, or planned marketing campaigns that have not yet happened. A forecast

built purely on historical patterns will tend to underreact to a genuinely new event, whether that is a downturn, a new competitor, or a deliberate promotional push the business itself decides to run.

The practical implication is that this system should be treated as a strong starting point for planning conversations, not a final answer that removes the need for judgment. A natural next step would be to enrich the model with known future events, such as planned promotions or pricing changes, and to feed in external indicators like broader retail spending trends, so that the forecast reflects not just what has happened before, but what is already known to be coming. As more years of data accumulate, the system's predictions and its ability to flag genuinely unusual weeks will also continue to improve.